TIER_1_SUMMARY

Tier 1 CLI Agents - Analysis Summary

> **10 Tier 1 agents analyzed** All agents are top-tier, production-ready CLI coding assistants

Agents Analyzed

#	Agent	Language	LOC	Key Differentiator
1	**claude-code-source**	TypeScript	513K	Internal features (KAIROS, Dream, Teams)
2	**aider**	Python	20K	Git-native pair programming
3	**codex**	TS/Rust	~10K	OpenAI official, sandboxed
4	**openhands**	Python	~50K	Autonomous SWE, evaluation
5	**cline**	TypeScript	~15K	Browser automation, VS Code
6	**continue**	TypeScript	~30K	Universal IDE support
7	**gemini-cli**	TypeScript	~5K	Google official, simple
8	**roo-code**	TypeScript	~15K	Multi-file editing
9	**swe-agent**	Python	~10K	SWE-bench focused
10	**vtcode**	Swift	~2K	Minimal, fast

Key Features Matrix

Feature	claude	aider	codex	oh	cline	cont	gem	roo	swe	vtc
Git Integration	✅	✅	✅	✅	✅	✅	✅	✅	✅	❌
Browser Automation	❌	❌	❌	✅	✅	❌	❌	❌	❌	❌
Sandbox/ Security	✅	❌	✅	✅	❌	❌	❌	❌	✅	❌
Voice Commands	❌	✅	❌	❌	❌	❌	❌	❌	❌	❌
Multi-file Edit	✅	✅	✅	✅	✅	✅	❌	✅	✅	❌
Repository Mapping	✅	✅	❌	✅	❌	✅	❌	❌	✅	❌
Evaluation Framework	❌	❌	❌	✅	❌	❌	❌	❌	✅	❌

Feature	claude	aider	codex	oh	cline	cont	gem	roo	swe	vtc
Team/ Swarm	✅	❌	❌	✅	❌	❌	❌	❌	❌	❌
Plan Mode	✅	❌	❌	❌	❌	❌	❌	❌	❌	❌
Auto-commit	❌	✅	✅	✅	✅	❌	❌	✅	❌	❌

Critical Features to Port to HelixAgent

1. From Claude Code Source

- **KAIROS**: Always-on background assistant
- **Dream System**: Memory consolidation with 3-gate triggers
- **Team Management**: Multi-agent swarms with consensus
- **Plan Mode**: Multi-step planning with verification gates
- **YOLO Classifier**: ML-based auto-approval
- **Permission System**: 4-layer permission architecture

2. From Aider

- **Repository Mapping**: Tree-sitter based codebase analysis
- **Edit Block Format**: Surgical code modifications
- **Git-Native Workflow**: Automatic commits, diff reviews
- **Voice Commands**: Speech-to-code integration
- **Lint/Test Integration**: Automatic quality checks

3. From Codex

- **Sandboxed Execution**: Secure command isolation
- **Approval System**: Multi-level approval gates
- **Protocol Architecture**: JSON-RPC communication
- **TUI**: Rich terminal UI with ratatui

4. From OpenHands

- **Agent System**: Pluggable agent architecture
- **Evaluation Framework**: SWE-bench integration
- **Event-Driven**: Action-observation loop
- **Runtime Environment**: Docker-based sandboxing
- **Browser Automation**: Headless browser integration

5. From Cline

- **Browser Integration**: Web navigation for research
- **Autonomous Execution**: Self-directed task completion
- **Context Management**: Smart file selection

6. From Continue

- **Context Providers**: Modular context system (@file, @url, @docs)

- **Action System**: Slash command framework
- **Universal IDE**: Works across editors

7. From Roo Code

- **Multi-file Editing**: Simultaneous file modifications
- **Context Optimization**: Token management
- **Agent Modes**: Specialized behaviors

8. From SWE-agent

- **Issue Resolution**: GitHub issue fixing
- **Computer Interface**: Filesystem + terminal
- **Thought-Action Loop**: Reasoning before acting

Implementation Priority for HelixAgent

Phase 1: Foundation (Already Done ✅)

- [x] Tool system with 30+ tools
- [x] Permission system with rules
- [x] Plan Mode with verification
- [x] Team management
- [x] KAIROS service
- [x] Dream system

Phase 2: Advanced Features (Next)

- [ ] Repository mapping with tree-sitter
- [ ] Edit block format
- [ ] Sandboxed execution
- [ ] Browser automation
- [ ] Voice commands
- [ ] Auto-commit workflow

Phase 3: Integration (Future)

- [ ] Context providers
- [ ] Action system
- [ ] Evaluation framework
- [ ] SWE-bench integration
- [ ] Agent modes

Feature Count by Agent

Agent	Total Features	Unique Features
claude-code-source	25	8 (KAIROS, Dream, Teams, etc.)
aider	18	4 (Voice, Repomap, etc.)
codex	12	3 (Sandbox, Protocol)
openhands	20	5 (Evaluation, Event system)
cline	14	2 (Browser automation)
continue	15	3 (Context providers)
roo-code	12	2 (Multi-file edit)
swe-agent	10	2 (SWE-bench)
gemini-cli	6	0
vtcode	4	0

Total Estimated Work

- **Features to port**: ~50 unique capabilities
- **Already implemented**: ~15 (30%)
- **Remaining**: ~35 features (70%)
- **Estimated time**: 4-6 weeks for full porting

Next Steps

1. ✅ Complete Tier 1 analysis
2. ⏳ Analyze Tier 2 agents
3. ⏳ Analyze Tier 3-5 agents
4. ⏳ Create master integration plan
5. ⏳ Implement features in priority order